

AMC 10 Miscellaneous Problems

Complete Solutions

Problem + Process + Final Answer

Problem 1

Problem. How many ways are there to color a 3×3 board using two colors, one color per square? Rotations and reflections are treated as different.

Solution. Each of the 9 squares can independently be colored in either of 2 colors. Therefore the multiplication principle gives 2^9 colorings.

Answer. 512

Problem 2

Problem. How many ways are there to color a 3×3 board using three colors if (a) every row contains all three colors; (b) every row and every column contains all three colors? Rotations and reflections are treated as different.

Solution. (a) Each row must be a permutation of the three colors, so each row has $3! = 6$ choices. The three rows are independent, giving $6^3 = 216$.

(b) Such a board is a 3×3 Latin square. Choose the first row in $3! = 6$ ways. Once the first row is fixed, the second row must be one of the two derangements of it; then the third row is forced. Thus there are $6 \cdot 2 = 12$ boards.

Answer. (a) 216; (b) 12

Problem 3

Problem. Sam walks around the outside of a regular pentagonal building with side length 30 m. He can see only 3 m around him. Find the total area of the region he can see during the walk.

Solution. The perimeter is $5 \cdot 30 = 150$ m. The visible exterior region is the 3-meter outer parallel band around the pentagon. Along the five sides, the band contributes $150 \cdot 3 = 450 \text{ m}^2$. At the vertices, the exterior circular sectors together have total angle 2π , so they contribute $\pi \cdot 3^2 = 9\pi \text{ m}^2$.

Answer. $450 + 9\pi$ square meters

Problem 4

Problem. Anna walks 1 mile north, turns 90° , walks 2 miles, turns 90° , walks 3 miles, turns 90° , and walks 1 mile. Find all possible values of AB.

Solution. At each turn she may turn left or right. Enumerating the eight turn patterns gives displacement vectors, up to signs and interchange of coordinates, equal to $(1,2)$, $(3,2)$, $(1,4)$, and $(3,4)$. Their lengths are $\sqrt{(1^2+2^2)}$, $\sqrt{(3^2+2^2)}$, $\sqrt{(1^2+4^2)}$, and $\sqrt{(3^2+4^2)}$.

Answer. $\sqrt{5}$, $\sqrt{13}$, $\sqrt{17}$, and 5 miles

Problem 5

Problem. If $a \oplus b = ab - a - b + 1$, find $1 \oplus (1 \oplus 2 + 2 \oplus 3 + 3 \oplus 4)$.

Solution. Notice that $a \oplus b = (a-1)(b-1)$. Therefore $1 \oplus 2 = 0$, $2 \oplus 3 = 2$, and $3 \oplus 4 = 6$. The quantity inside the outer operation is $0+2+6=8$. Hence $1 \oplus 8 = (1-1)(8-1)=0$.

Answer. 0

Problem 6

Problem. How many ways are there to color the six faces of a cube using two colors, where colorings that differ only by a rotation are identical?

Solution. Classify by the number of faces of the first color. For 0, 1, 5, or 6 such faces, there is 1 pattern each. For 2 faces, the two faces are either adjacent or opposite, giving 2 patterns. By color-complement symmetry, 4 faces also gives 2 patterns. For 3 faces, either some pair is opposite or all three meet at one vertex, giving 2 patterns. Thus the total is $1+1+2+2+2+1+1$.

Answer. 10

Problem 7

Problem. Find the largest integer x such that $x^2 > 100$ and $x^3 < 64$.

Solution. From $x^2 > 100$, either $x > 10$ or $x < -10$. From $x^3 < 64$, we have $x < 4$. The first branch is impossible, so $x < -10$. The largest integer satisfying this is -11 .

Answer. -11

Problem 8

Problem. Find the area of the quadrilateral with vertices $(0,1)$, $(3,5)$, $(6,2)$, and $(5,0)$.

Solution. Use the shoelace formula in the given cyclic order:

$$\begin{aligned} \text{Area} &= 1/2 |(0 \cdot 5 + 3 \cdot 2 + 6 \cdot 0 + 5 \cdot 1) - (1 \cdot 3 + 5 \cdot 6 + 2 \cdot 5 + 0 \cdot 0)| \\ &= 1/2 |11 - 43| = 16. \end{aligned}$$

Answer. 16 square units

Problem 9

Problem. Find the area of the triangle formed by $y = 3x + 4$, $y = 2x - 4$, and $y = -2x + 4$.

Solution. Find the intersections. The first two lines meet at $(-8, -20)$. The first and third meet at $(0, 4)$. The second and third meet at $(2, 0)$. Using the shoelace formula on these three points gives
$$\text{Area} = 1/2 |(-8 \cdot 4 + 0 \cdot 0 + 2 \cdot (-20)) - ((-20) \cdot 0 + 4 \cdot 2 + 0 \cdot (-8))| = 40.$$

Answer. 40 square units

Problem 10

Problem. Compute $\sqrt{9 - 6\sqrt{2}} + \sqrt{9 + 6\sqrt{2}}$.

Solution. Since $(\sqrt{6} - \sqrt{3})^2 = 9 - 6\sqrt{2}$ and $(\sqrt{6} + \sqrt{3})^2 = 9 + 6\sqrt{2}$, both radicals simplify to positive quantities. Their sum is $(\sqrt{6} - \sqrt{3}) + (\sqrt{6} + \sqrt{3}) = 2\sqrt{6}$.

Answer. $2\sqrt{6}$

Problem 11

Problem. A disk of radius 1 rolls inside a circle of radius 2. Initially a point P on the disk touches the larger circle. Find the total distance traveled by P until it touches the larger circle again.

Solution. The path of a point on a circle rolling inside a circle of twice its radius is a diameter of the larger circle. Starting at one endpoint, P moves along the diameter to the opposite endpoint, where it next touches the larger circle. The diameter is $2 \cdot 2 = 4$.

Answer. 4 units

Problem 12

Problem. Find the coefficient of x^3 in $(x^2 + 1/x)^{15}$.

Solution. If j factors of $1/x$ are chosen, then $15-j$ factors of x^2 are chosen. The resulting exponent is $2(15-j)-j = 30-3j$. Set $30-3j=3$, so $j=9$. The coefficient is $C(15,9)=5005$.

Answer. 5005

Problem 13

Problem. Find 8 lattice points that are 5 units from $(13, 71)$.

Solution. Integer offsets (u, v) satisfying $u^2 + v^2 = 25$ include $(\pm 5, 0)$, $(0, \pm 5)$, and $(\pm 3, \pm 4)$. Any eight valid translated points may be used. For example, translating the eight offsets $(\pm 5, 0)$, $(0, \pm 5)$, and $(\pm 3, 4)$ gives the listed points.

Answer. $(18, 71)$, $(8, 71)$, $(13, 76)$, $(13, 66)$, $(16, 75)$, $(10, 75)$, $(16, 67)$, $(10, 67)$

Problem 14

Problem. What is the probability that 3 randomly drawn cards from a standard 52-card deck are all of the same suit?

Solution. There are $C(52,3)$ unordered 3-card hands. For a favorable hand, choose the suit in 4 ways and then choose 3 of its 13 cards: $4C(13,3)$. Thus
 $P = 4C(13,3)/C(52,3) = 4 \cdot 286/22100 = 22/425$.

Answer. 22/425

Problem 15

Problem. Find the number of trailing zeros of $100!$ when written in base 12.

Solution. Since $12=2^2 \cdot 3$, each trailing base-12 zero requires two factors of 2 and one factor of 3. Legendre's formula gives

$$v_2(100!) = 50 + 25 + 12 + 6 + 3 + 1 = 97,$$

$$v_3(100!) = 33 + 11 + 3 + 1 = 48.$$

Therefore the number of zeros is $\min(\lfloor 97/2 \rfloor, 48) = 48$.

Answer. 48

Problem 16

Problem. Find the smallest positive integer k such that no factorial $n!$ has exactly k trailing zeros when written in base 9.

Solution. Because $9=3^2$, the number of trailing base-9 zeros of $n!$ is $\lfloor v_3(n!)/2 \rfloor$. These values increase only when n passes a multiple of 3. Checking the values up to the first jump that skips an integer: for $n=24$, $v_3(24!)=8+2=10$, giving 5 zeros; continuing through $n=26$ gives 5, $n=27$ gives $v_3(27!)=9+3+1=13$ and hence 6 zeros. Continuing similarly, all values 1 through 11 occur, but the jump at $n=54$ is from 11 to 13 because $v_3(53!)=22$ and $v_3(54!)=25$. Therefore 12 is skipped.

Answer. 12

Problem 17

Problem. Let $p(x) = \prod_{n=0}^{100} (x-n)^n$. (a) Find the number of digits in $p(37)$. (b) Find $p(600!) \bmod 97$. (c) Find the coefficient of the second-highest-degree term of $p(x)$.

Solution. (a) The factor $(x-37)^{37}$ occurs, so $p(37)=0$. The number 0 has one digit.

(b) Since 97 divides $600!$, we have $600! \equiv 0 \pmod{97}$. The product also contains the factor $(x-97)^{97}$, which is 0 modulo 97 at $x=600!$. Hence the remainder is 0.

(c) The degree is $\sum_{n=1}^{100} n = 5050$. In a monic polynomial, the coefficient of x^{5049} is the negative of the sum of all roots counted with multiplicity. Root n has multiplicity n , so the coefficient is $-\sum_{n=1}^{100} n^2 = -100 \cdot 101 \cdot 201/6 = -338350$.

Answer. (a) 1 digit; (b) 0; (c) -338350

Problem 18

Problem. A king starts at the bottom-left cell of a 5×5 board and may move one cell up, right, or diagonally up-right. How many paths lead to the top-right cell?

Solution. The king must change its coordinates by $(4,4)$. If it makes k diagonal moves, it then needs $4-k$ up moves and $4-k$ right moves. The number of paths for that k is $(8-k)!/[k!(4-k)!(4-k)!]$. Summing for $k=0,1,2,3,4$ gives

$$70 + 140 + 90 + 20 + 1 = 321.$$

Answer. 321

Problem 19

Problem. Find $\gcd(20727036, 20727432)$.

Solution. Use the Euclidean algorithm. Their difference is 396, so $\gcd(20727036, 20727432) = \gcd(20727036, 396)$.

Since $20727036 = 396 \cdot 52341$, the remainder is 0.

Answer. 396

Problem 20

Problem. Find the coefficient of x^{999} in $(x+1)(x+2)\cdots(x+1000)$.

Solution. The product is monic of degree 1000. To obtain x^{999} , choose the constant term from exactly one factor and x from all the others. Therefore the coefficient is $1+2+\cdots+1000 = 1000 \cdot 1001/2$.

Answer. 500500

Problem 21

Problem. Find the area of a rhombus with side length 3 if one angle is twice another.

Solution. Adjacent angles of a rhombus are supplementary. Let the smaller angle be θ ; then the larger is 2θ , so $\theta+2\theta=180^\circ$, giving $\theta=60^\circ$. The area is $s^2 \sin \theta = 3^2 \sin 60^\circ = 9\sqrt{3}/2$.

Answer. $9\sqrt{3}/2$ square units

Problem 22

Problem. Find the last four digits of 101^{1000} .

Solution. Write $101=1+100$. By the binomial theorem,
 $(1+100)^{1000} = 1 + 1000 \cdot 100 + \text{terms containing } 100^2$.

The second term is 100000, divisible by 10000, and every later term is also divisible by 10000. Thus the number is congruent to 1 modulo 10000.

Answer. 0001

Problem 23

Problem. Find a row of Pascal's triangle containing at least 8 entries divisible by 3.

Solution. Take row $n=9$, whose entries are $C(9,0), C(9,1), \dots, C(9,9)$:
1, 9, 36, 84, 126, 126, 84, 36, 9, 1.

The eight interior entries are all divisible by 3.

Answer. Row $n=9$ (the 10th row if the top row is called row 1)

Problem 24

Problem. Find the sum of the coefficients of $(2xy + z - 3)^{100}$.

Solution. For a multivariable polynomial, the sum of all coefficients is obtained by setting every variable equal to 1. Thus the sum is $(2 \cdot 1 \cdot 1 + 1 - 3)^{100} = 0^{100} = 0$.

Answer. 0

Problem 25

Problem. Find the number of trailing zeros of $C(1000, 500)$.

Solution. The number of trailing decimal zeros is $\min(v_2, v_5)$. Using Legendre's formula on the binomial coefficient:

$$v_5 = v_5(1000!) - 2v_5(500!) = (200 + 40 + 8 + 1) - 2(100 + 20 + 4) = 249 - 248 = 1.$$

Also $v_2 = 6$, so the limiting prime is 5.

Answer. 1

Problem 26

Problem. Anna and Bob play until one player leads by 2 points. Anna is twice as likely as Bob to score each point. Find the probability Anna wins.

Solution. Let $p=2/3$ and $q=1/3$. Let u_0 be Anna's winning probability from a tie, u_1 from a one-point lead, and u_{-1} from a one-point deficit. Then

$$u_1 = p + qu_0, \quad u_{-1} = pu_0, \quad \text{and} \quad u_0 = pu_1 + qu_{-1}.$$

Substitution gives $u_0 = p^2 + 2pqu_0$, so

$$u_0 = p^2 / (1 - 2pq) = (4/9) / (5/9) = 4/5.$$

Answer. 4/5

Problem 27

Problem. Find a formula for the area of a regular octagon with side length s .

Solution. Divide the octagon into 8 congruent isosceles triangles from the center. Using the standard regular-polygon formula $A = ns^2/[4\tan(\pi/n)]$ with $n=8$,

$$A = 8s^2/[4\tan(\pi/8)] = 2s^2/(\sqrt{2}-1) = 2(1+\sqrt{2})s^2.$$

Answer. $2(1+\sqrt{2})s^2$

Problem 28

Problem. If 5 men can dig 10 wells in 20 days, how many men are needed to dig 15 wells in 8 days?

Solution. The first job uses $5 \cdot 20 = 100$ man-days for 10 wells, or 10 man-days per well. Fifteen wells require 150 man-days. Over 8 days, the required workforce is $150/8 = 75/4 = 18.75$ men. Since the number of workers must be whole, 19 men are needed to finish within 8 days.

Answer. 19 men (the proportional rate is 18.75)

Problem 29

Problem. Anna, Bob, Cat, and Denis distribute 24 identical pieces of candy. Each gets at least 2 pieces, and Anna gets at least as much as Bob. How many distributions are possible?

Solution. Subtract 2 pieces from each child. Let $a, b, c, d \geq 0$ with $a+b+c+d=16$ and $a \geq b$. The total number without the inequality is $C(19,3)=969$. By symmetry, the number with $a > b$ equals the number with $b > a$. Count the cases $a=b$: if $a=b=t$, then $c+d=16-2t$, so the number is $\sum$ from $t=0$ to 8 of $(17-2t)=81$. Therefore the desired number is $(969+81)/2=525$.

Answer. 525

Problem 30

Problem. Three pizzas are cut into 24 identical slices. In how many ways can six distinct children divide the slices if each gets at least 1 and at most 5 slices?

Solution. Let x_i be the number received by child i . We need $x_1 + \dots + x_6 = 24$ with $1 \leq x_i \leq 5$. Set $y_i = 5 - x_i$. Then $0 \leq y_i \leq 4$ and $y_1 + \dots + y_6 = 6$. Without the upper bounds there are $C(11,5)=462$ solutions. Subtract solutions in which some $y_i \geq 5$: choose that child in 6 ways and count $C(6,5)=6$ remaining solutions. Two variables cannot both be at least 5 because their sum would exceed 6. Thus the answer is $462 - 6 \cdot 6 = 426$.

Answer. 426